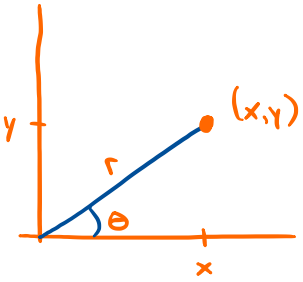
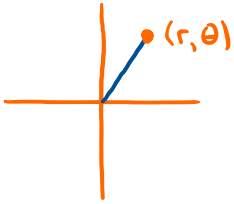
8.3 Polar coordinates

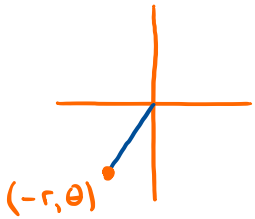
A point (x,y) in the plane has polar coordinates (r,θ) where:



$$x = r\cos(\theta)$$
$$y = r\sin(\theta)$$
$$x^2+y^2=r^2$$
$$\tan(\theta) = y/x$$

We allow r to be negative — a negative r just means the point is $|r|$ units away from the origin in the opposite direction of θ. For example:





Thus $(-r,\theta) = (r, \theta+\pi)$. We also allow θ to be any angle, not just in $[0,2\pi)$. The usual coordinates are called cartesian or rectangular.

→ Do all problems.